PBI Lonza

Lonza uses **Microsoft Power BI (PBI)** and Microsoft Azure together as an integrated, cloud-based data analytics and operational intelligence platform. This technology stack enables them to optimize the global manufacturing of biologic drugs, cell therapies, and small-molecule medicines. [1, 2, 3, 4]

How Lonza Uses Azure and Data

- **Operational Data Foundations:** Lonza's Process Analytics & Data Intelligence (PADI) teams use Azure to connect disparate data sources—such as manufacturing execution systems (MES), laboratory notebooks, and plant historians—into a coherent, scalable operational data foundation. [1]
- **Digital Twins & Predictive Modeling:** Utilizing Azure's cloud infrastructure, Lonza leverages AI-enabled tools and platforms like *Design2Optimize* to create "digital twins" of active pharmaceutical ingredient (API) manufacturing processes. This allows scientists to run scenario tests virtually to speed up development. [1, 2]
- **Preclinical Development:** Lonza uses Azure-powered compute and storage to run Physiologically Based Biopharmaceutics Modeling (PBBM). This integrates biochemical data to predict how a drug distributes, metabolizes, and eliminates in humans to de-risk assets. [1]
- **Global Scalability:** Moving process and asset data to a scalable Azure infrastructure allows Lonza to tackle operational pain points globally and seamlessly integrate new manufacturing sites as the business expands. [1]

How Lonza Uses Power BI

- **Data Visualization & Dashboards:** Power BI is used across Lonza to create interactive, real-time data visualizations. These are heavily relied upon by site operators, Process Development, and MSAT (Manufacturing Science and Technology) teams to monitor ongoing production runs. [1, 2]
- **Process Optimization:** By pulling data from Azure into Power BI, development teams can gain near-real-time visibility into production performance, allowing them to track yields, troubleshoot inefficiencies, and enhance process robustness. [1, 2, 3]

Would you like to know more about how they integrate specific data streams like **OSIsoft PI historians** or how they secure their cloud infrastructure for **GxP-regulated pharmaceutical workloads**?

OSIsoft PI historians

An **OSIsoft PI System** (now part of AVEVA) is a specialized industry-standard data management platform designed to collect, store, and analyze massive volumes of real-time time-series data from physical equipment, manufacturing plants, and sensors. [1, 2, 3, 4, 5]

It is widely considered the gold standard "historian" in heavily regulated, asset-intensive industries like pharmaceutical manufacturing, energy, and utilities. [1]

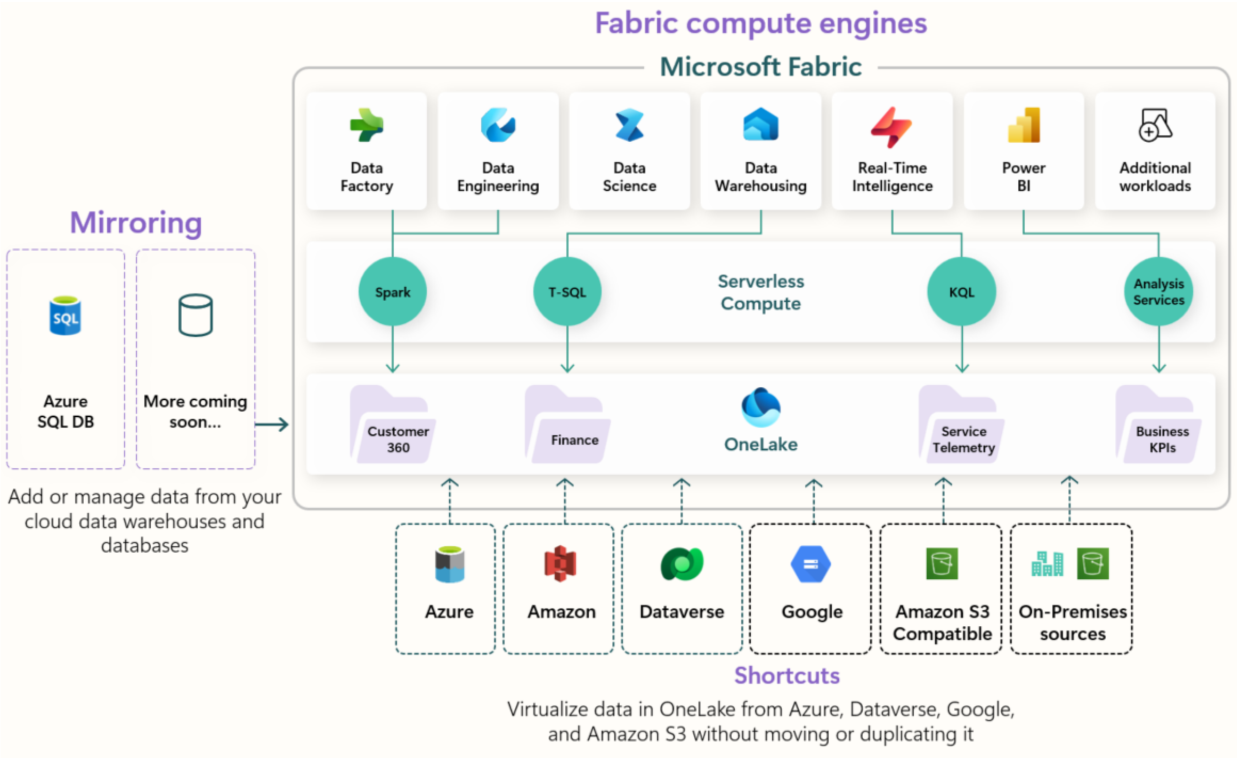
Key Capabilities

- **High-Speed Data Collection:** Connects directly to thousands of diverse plant floor assets, automation systems, and lab equipment via native industrial protocols (like OPC, Modbus, or MQTT). [1]
- **Massive Time-Series Storage:** Compresses and stores sub-second data points—such as temperature, pressure, pH levels, and flow rates—for decades without losing data fidelity. [1, 2, 3]
- **Contextualization:** Uses a feature called **Asset Framework (AF)** to organize raw sensor data into an intuitive hierarchy that mirrors physical equipment (e.g., *Site > Building > Bioreactor 3 > Temperature Sensor*). [1]
- **Event Tracking:** Features **Event Frames** to automatically bookmark specific operational windows, such as a precise 12-hour pharmaceutical batch run, so engineers can easily compare past runs to find anomalies.

Why Industries Like Pharma Use It

- **Audit and Compliance:** Provides an unalterable, high-fidelity audit trail required by regulatory bodies like the FDA to prove a medicine was manufactured under exact specified conditions (GxP compliance).
- **Operational Visibility:** Feeds real-time data directly into cloud platforms like Microsoft Azure or visualization tools like Power BI, bridging the gap between factory-floor machines and corporate IT analytics. [1, 2]

Would you like to explore **how PI data is exported to Azure** for advanced analytics, or should we look at how engineers use **PI Event Frames to monitor pharmaceutical batches**?



OneLake, **Lakehouse**, Dataverse, and **Cosmos DB** are all Microsoft data platforms, but they serve fundamentally different roles in a data ecosystem: **OneLake** is the storage foundation, **Lakehouse** is the analytics engine, **Dataverse** is a business application database, and **Cosmos DB** is an operational NoSQL database. [1, 2, 3, 4, 5]

1. OneLake

OneLake is the foundational, tenant-wide data lake included with every Microsoft Fabric deployment. [1, 2]

- **Core Purpose:** Acts as the "OneDrive for your organization's data," centralizing storage to eliminate data silos.
- **Structure:** Stores all tabular data uniformly in open-source Delta Parquet format.
- **Key Feature:** Uses **shortcuts** and **mirroring** to virtualize data or bring operational data in near real-time without physical duplication.
- **Best Used For:** Centralizing global governance, security, and storage for all enterprise data. [1, 2, 3]

2. Lakehouse

The Lakehouse is an analytics workload within Microsoft Fabric. It combines the flexibility of a traditional data lake with the structure of a data warehouse. [1, 2, 3, 4]

- **Core Purpose:** Processing, combining, and querying vast amounts of both structured and unstructured big data.

- **Structure:** It acts as a compute interface on top of OneLake, offering both Spark compute (for data engineering) and a SQL analytics endpoint (for reporting).
- **Key Feature:** Integrates seamlessly with **Power BI** for fast analytics using Direct Lake mode.
- **Best Used For:** Machine learning workflows, data science, and heavy big-data transformations. [1, 2, 3, 4, 5, 6]

3. Dataverse

Dataverse is a low-code, relational, cloud-based data platform primarily tied to Microsoft Dynamics 365 and Power Apps. [1, 2, 3, 4, 5]

- **Core Purpose:** The central repository for business applications to safely store, manage, and secure business logic.
- **Structure:** It is more than just a raw database; it features built-in security, roles, relationships, and business rules.
- **Key Feature:** Connects directly to Microsoft Fabric, automatically syncing its data into OneLake for analytics.
- **Best Used For:** Building operational business applications, CRM deployments, and transactional app data. [1, 2, 3, 4, 5]

4. Cosmos DB

Azure Cosmos DB is a globally distributed, NoSQL (and multimodel) operational database. [1, 2, 3, 4]

- **Core Purpose:** Powering high-throughput, mission-critical applications that require sub-millisecond response times.
- **Structure:** Designed to store semi-structured, JSON, and highly scalable data.
- **Key Feature:** Features **zero-ETL integration** with Microsoft Fabric, allowing you to mirror your Cosmos DB operational data directly into OneLake automatically.
- **Best Used For:** Real-time application serving, IoT applications, and massive-scale transactional workloads. [1, 2, 3, 4, 5]

Quick Summary

Technology [1, 2, 3, 4, 5, 6, 7, 8, 9]	Primary Role	Data Type	Typically Found In
OneLake	Storage & Governance	Any files & Delta tables	Microsoft Fabric
Lakehouse	Analytics & ML	Unstructured & Structured	Microsoft Fabric

Dataverse	Business Applications	Relational / Business	Power Platform / Dynamics 365
Cosmos DB	App Transactions / NoSQL	Semi-structured / JSON	Azure / Microsoft Fabric

Nexus

Lonza is undergoing a multi-year global digital transformation program called "Nexus," which involves deploying the SAP S/4HANA platform. This initiative harmonizes and manages master data, supply chain logistics, and manufacturing across its global CDMO (Contract Development and Manufacturing Organization) sites. [1, 2, 3, 4]

Lonza leverages several specific components of SAP Data and ERP systems to support its biotech operations:

- **SAP S/4HANA:** The core enterprise resource planning (ERP) system used to connect global operations, integrating supply chain management, Quality Management (QM), and Production Planning (PP).
- SAP Master Data Governance (MDG)**:** Used to define, standardize, and maintain uniform master data rules across various disparate legacy systems before migrating them to the central S/4HANA infrastructure. [1]
- SAP Business Technology Platform (BTP)**:** Employed by Lonza to build and extend the capabilities of its central ERP, utilizing BTP's analytics and data management tools. [1]
- SAP Business Network**:** Adopted globally to manage workflows with suppliers, process purchase orders, and strengthen supply chain resilienc

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